Math 225A
Topics in Number Theory

Nayeong Kim
Lectured by Prof. Amalendu Krishna

Introduction

This note is based on the lecture of Professor Krishna, and written by Nayeong Kim. In this course, we will explore the following topics.

- Galois Cohomology
- Central Simple Algebras
- Brauer Groups and their Representation with Galois Cohomology
- Brauer Groups of Local and Global Fields

AI Disclaimer

I am using Claude for maintaining the notes in the git repository and for generating formatting and some Latex code.

Chapter 1

Galois Cohomology

1.1 Group Cohomology

Let G be a discrete group, and let $p\colon EG \to BG$ be the universal principal G-bundle over the classifying space BG. Its defining property is that, for every paracompact Hausdorff space X, pulling back along maps $X \to BG$ induces a bijection

$$[X, BG] \xrightarrow{\cong} \mathrm{Prin}_G(X)$$

between homotopy classes of maps $X \to BG$ and isomorphism classes of principal G-bundles over X.

In particular, let $\pi\colon P \to X$ be a principal G-bundle. Then there is a map $f\colon X \to BG$, unique up to homotopy, and a G-equivariant map $\tilde{f}\colon V \to EG$ such that the square

$$\begin{array}{ccc} V & \xrightarrow{\tilde{f}} & EG \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle p} \\ X & \xrightarrow[f]{} & BG \end{array}$$

is a pullback; that is, $V \cong f^*EG$ as principal G-bundles.

We note that since G is discrete, we have $\Pi_1(BG) \cong G$ and $\Pi_i(BG) = 0$ for $i > 1$.

So BG is a $K(G,1)$ space, and we have

$$H^i(G,M) \cong H^i(BG,M)$$

for any G-module M.

Now, when we take a look at $H^*_{sing}(BG,\mathbb{Z})$, it is much more complicated than BG but it gives us a lot of information about G. This gives a geometric motivation for group cohomology as we study representations of G to understand G better.

We can study it when we want to understand G, the Galois group of some field. For example, when $G = \mathbb{Z}$, we have $BG = S^1$ and $EG = \mathbb{R}$ and we can compute $H^*(S^1,\mathbb{Z})$. For another example, when $G = \mathbb{Z}/2\mathbb{Z}$, we have $BG = \mathbb{R}P^\infty$ and $EG = S^\infty$ and we can compute $H^*(\mathbb{R}P^\infty,\mathbb{Z})$. More generally, when $G = \mathbb{Z}/n\mathbb{Z}$, we have $BG = L^\infty_n := S^\infty/(\mathbb{Z}/n\mathbb{Z})$, the infinite lens space, and $EG = S^\infty$. Here we regard S^∞ as the unit sphere in $\mathbb{C}^\infty = \bigcup_k \mathbb{C}^k$, and a generator of $\mathbb{Z}/n\mathbb{Z}$ acts by multiplication by $\zeta = e^{2\pi i/n}$ on each coordinate. This action is free, and S^∞ is contractible, so $S^\infty \to L^\infty_n$ is a universal principal $\mathbb{Z}/n\mathbb{Z}$-bundle. Using the CW structure on L^∞_n with one cell in each dimension, whose cellular chain complex is

$$\mathbb{Z} \xleftarrow{0} \mathbb{Z} \xleftarrow{n} \mathbb{Z} \xleftarrow{0} \mathbb{Z} \xleftarrow{n} \mathbb{Z} \xleftarrow{0} \cdots,$$

we obtain

$$H^k(L^\infty_n,\mathbb{Z}) \cong \begin{cases} \mathbb{Z} & k = 0, \\ \mathbb{Z}/n\mathbb{Z} & k > 0 \text{ even}, \\ 0 & k \text{ odd}. \end{cases}$$

In fact, as a ring, $H^*(L^\infty_n,\mathbb{Z}) \cong \mathbb{Z}[x]/(nx)$ with $\deg x = 2$. Setting $n = 2$ recovers $H^*(\mathbb{R}P^\infty,\mathbb{Z})$, since $L^\infty_2 = \mathbb{R}P^\infty$.

To study such objects, we cannot find a good answer with topological methods, so we need to use algebraic methods. We can study cohomology groups and Galois cohomology groups with algebraic methods.

What is a group cohomology? For simplicity, we assume that G is a finite group (although the profiniteness is enough). Let A be an abelian group with a G-action,

i.e., a G-module. Let G-Mod be the category of G-modules and G-equivariant homomorphisms. We note that G-Mod is an abelian category with enough injectives.

Let's consider an example. Let $\mathbb{Z}[G]$ be the group algebra of G over $\mathbb{Z}$. In fact, this is a canonical way of constructing a ring from a group. Here are some properties of $\mathbb{Z}[G]$.

Remark 1.

1. $\mathbb{Z}[G]$ is commutative if and only if G is abelian.
2. $\mathbb{Z}[G]$ is noetherian if and only if G is finitely generated.

Remark 2. The algebra $\mathbb{Z}[G]$ is an augmented $\mathbb{Z}$-algebra with an augmentation $\epsilon\colon \mathbb{Z}[G] \to \mathbb{Z}$ map given by

$$\sum_{g\in G} m_g g \longmapsto \sum_{g\in G} m_g.$$

The augmentation ideal I_G is defined as the kernel of ϵ so it is clear that it is a 2-sided ideal of $\mathbb{Z}[G]$.

We also note that $\mathbb{Z}[G]$ is a free $\mathbb{Z}$-module with a $\mathbb{Z}$-basis G. So is I_G with the following lemma.

Lemma 1. The ideal I_G has a $\mathbb{Z}$-basis $\{g-1 \mid g \in G, g \neq 1\}$.

Proof. Let $g \in G$ with $g \neq 1$. Then we have

$$\epsilon(g-1) = \epsilon(g) - \epsilon(1) = 1 - 1 = 0,$$

so $g - 1 \in I_G$. Now, let $\sum_{g\in G} m_g g \in I_G$. Then we have

$$0 = \epsilon\left(\sum_{g\in G} m_g g\right) = \sum_{g\in G} m_g.$$

So we can write

$$\sum_{g\in G} m_g g = m_1 + \sum_{g\neq 1} m_g g = -\sum_{g\neq 1} m_g + \sum_{g\neq 1} m_g g = \sum_{g\neq 1} m_g(g-1).$$

This shows that $\{g-1\}_{g\neq 1}$ generates I_G as a $\mathbb{Z}$-module.

To show that it is linearly independent, suppose that $\sum_{g\neq 1} m_g(g-1)=0$. Then we have $\sum_{g\neq 1} m_g g + \sum_{g\neq 1}(-m_g)1 = 0$. Since the elements of G are linearly independent in $\mathbb{Z}[G]$, we must have $m_g = 0$ for all $g \neq 1$. □

The action of G is invariant on I_G since $h\cdot(g-1) = hg-h = (hg-1)-(h-1) \in I_G$ for all $g, h \in G$. So we have $\mathbb{Z}[G] = \mathbb{Z} \oplus I_G$ as G-modules.

As we mentioned, we can consider a ring $\mathbb{Z}[G]$ canonically from a group G. Not surprisingly, we can consider a G-module A as a $\mathbb{Z}[G]$-module via $(\sum_{g\in G} m_g g)\cdot a = \sum_{g\in G} m_g(g\cdot a)$ for $a \in A$. The converse is also true. There is a composition of homomorphism $G \to \mathbb{Z}[G]^\times \to \mathrm{Aut}_{ab}(A)$ which gives a canonical G-action on A.

Proposition 1. There is an equivalence of abelian categories between G-Mod and $\mathbb{Z}[G]$-Mod.

Now we consider a functor $F : G\text{-}\mathsf{Mod} \to \mathsf{Ab}$ which sends A to $A^G = \{a \in A \mid g\cdot a = a \text{ for all } g \in G\}$. We may consider an abelian group A as a G-module with a trivial action so we have a functor $\mathrm{tr} : \mathsf{Ab} \to G\text{-}\mathsf{Mod}$.

Claim 1. We claim that (tr, F) is an adjoint pair of functors.

Proof. For any $A \in \mathsf{Ab}$ and $B \in G\text{-}\mathsf{Mod}$, we have a natural isomorphism

$$\mathrm{Hom}_{G\text{-}\mathsf{Mod}}(\mathrm{tr}(A), B) \cong \mathrm{Hom}_{\mathsf{Ab}}(A, B^G).$$

For any G-equivariant homomorphism $\phi : \mathrm{tr}(A) \to B$, we have a homomorphism $\psi : A \to B^G$ given by

$$\psi(a) = \phi(a) \in B^G.$$

Conversely, for any homomorphism $\psi : A \to B^G$, we have a G-equivariant homomorphism $\phi : \mathrm{tr}(A) \to B$ given by

$$\phi(a) = \psi(a) \in B.$$

It is clear that these two maps are inverses of each other. □

With the claim, we have a corollary.

Corollary 1. The invariant functor $F : G\text{-}\mathsf{Mod} \to \mathsf{Ab}$ is left exact.

But note that F is not right exact in general. For example, let $G = \mathbb{Z}/2\mathbb{Z}$ and take the augmentation sequence $0 \to I_G \to \mathbb{Z}[G] \to \mathbb{Z} \to 0$ which is exact with the definition of I_G. But $0 \to I_G^G = 0 \to \mathbb{Z}[G]^G = \mathbb{Z} \to \mathbb{Z}^G = \mathbb{Z} \to 0$ is not exact.

With the left exactness of F, we can consider the right derived functor R^iF. But we need to be careful; we need to have enough injectives in $G\text{-}\mathsf{Mod}$ to define the derived functor.

Lemma 2. Let A be an abelian group. Then there exists an injective abelian group I and an injective homomorphism $A \to I$.

Proof. Let S be the set of all abelian group homomorphisms from A to $\mathbb{Q}/\mathbb{Z}$. Let I be S-copies of $\mathbb{Q}/\mathbb{Z}$. It is clear that I is injective since $\mathbb{Q}/\mathbb{Z}$ is injective.

We have a homomorphism $\Phi : A \to I$ given by $a \mapsto (\phi(a))_{\phi \in S}$. We want to show that Φ is injective, and it suffices to show that for any nonzero $a \in A$, there exists a homomorphism $\phi : A \to \mathbb{Q}/\mathbb{Z}$ such that $\phi(a) \neq 0$.

Let $a \in A$ be nonzero. Then we have a subgroup $\langle a \rangle$ of A generated by a. We have a homomorphism $\psi : \langle a \rangle \to \mathbb{Q}/\mathbb{Z}$ given by $ma \mapsto m/n + \mathbb{Z}$ where n is the order of a. By the injectivity of $\mathbb{Q}/\mathbb{Z}$, we can extend ψ to a homomorphism $\phi : A \to \mathbb{Q}/\mathbb{Z}$. Then we have a non-zero homomorphism ϕ with $\phi(a) = \psi(a) = 1/n + \mathbb{Z} \neq 0$. It follows that Φ is injective. □

Proposition 2. Let R be a ring and M be an R-module. Then there are enough injective R-modules.

Proof. Let M be a left R-module. Look at the composition of R-module homomorphisms $M \xrightarrow{\sim} \mathrm{Hom}_R(R, M) \hookrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, M)$ where $M \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{Z}}(R, M)$ is given by $m \mapsto (r \mapsto rm)$. We take an injective module I as in the previous lemma. Then we have an injection $\mathrm{Hom}_{\mathbb{Z}}(R, M) \hookrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, I)$ via $M \hookrightarrow I$.

It is left to show that $I' := \mathrm{Hom}_{\mathbb{Z}}(R, I)$ is injective as an R-module. Take any injection $A \hookrightarrow B$ of R-modules. We want to show that $\mathrm{Hom}_R(B, I') \twoheadrightarrow \mathrm{Hom}_R(A, I')$.

We observe that $\mathrm{Hom}_R(B, I') = \mathrm{Hom}_R(B, \mathrm{Hom}_{\mathbb{Z}}(R, I)) \cong \mathrm{Hom}_{\mathbb{Z}}(B, I)$ via the adjointness of Hom functors. Similarly, we have $\mathrm{Hom}_R(A, I') \cong \mathrm{Hom}_{\mathbb{Z}}(A, I)$. Since I is injective as an abelian group (as a $\mathbb{Z}$-module), we have $\mathrm{Hom}_{\mathbb{Z}}(B, I) \twoheadrightarrow \mathrm{Hom}_{\mathbb{Z}}(A, I)$. It follows that $\mathrm{Hom}_R(B, I') \twoheadrightarrow \mathrm{Hom}_R(A, I')$ and I' is injective as an R-module. □

Now we work with a G-module A.

Lemma 3. $A^G = \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, A)$.

Proof. By adjointness, we have $\mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, A) \cong \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, A)^G = A^G$. □

1.2 Derived Categories

Since we have enough injective G-modules, we can consider the derived functor. We consider an injective resolution of A as a $\mathbb{Z}[G]$-module

$$0 \to A \to I^0 \to I^1 \to I^2 \to \cdots .$$

From the cochain complex, we have

$$\mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, A) = H^i(\mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, I)).$$

We note that it doesn't depend on the choice of injective resolution of A; hence it is well-defined.

Definition 1. Define the i-th cohomology group of G with coefficients in A as $H^i(G, A) := \mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, A)$.

Here is a corollary of the previous lemma.

Corollary 2. We have $H^0(G, A) = A^G$.

How do we actually compute $H^i(G, A)$? The hardness of the computation depends on the choice of the injective resolution of A. For $\mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, A)$, instead of taking an injective resolution of A, we can take a projective resolution of $\mathbb{Z}$ as a $\mathbb{Z}[G]$-module. One benefit of this approach is that projective resolutions are often easier to construct than injective resolutions since projective modules are direct summands of free modules.

Recall the following theorem.

Theorem 1. Let M be a projective R-module and N be an R-module. Then $\mathrm{Ext}^i_R(M, N) = 0$ for all $i > 0$.

Proof. We may assume that M is a free R-module since projective modules are direct summands of free modules. Let $M = \bigoplus_{j \in I} R$ be a free R-module. Then we have $\mathrm{Ext}^i_R(M, N) = \mathrm{Ext}^i_R\left(\bigoplus_{j \in I} R, N\right) = 0$ for all $i > 0$. □

Corollary 3. If $f : A \to B$ is a G-invariant map, then there is a canonical map $f^i : H^i(G, A) \to H^i(G, B)$ for all $i \geq 0$.

Corollary 4. $H^0(G, A) = A^G$, and $H^1(G, A)$ is the set of equivalence classes of extensions of $\mathbb{Z}$ by A as $\mathbb{Z}[G]$-modules.

Corollary 5. Let $0 \to A' \to A \to A'' \to 0$ be a short exact sequence of G-modules. Then we have a long exact sequence

$$\cdots \to H^i(G, A') \to H^i(G, A) \to H^i(G, A'') \to H^{i+1}(G, A') \to \cdots .$$

We are still working with a finite group G.

Definition 2. The *norm element* of $\mathbb{Z}[G]$ is defined as the sum of all elements in G:

$$N := \sum_{g \in G} g.$$

Remark 3.

1. The norm element N is a central element of $\mathbb{Z}[G]$ since we have

$$gN = g \sum_{h \in G} h = \sum_{h \in G} gh = N$$

for any $g \in G$. Likewise $Ng = N$. Hence $gN = N = Ng$.

2. N is a G-invariant element of $\mathbb{Z}[G]$ since we have

$$g \cdot N = g \cdot \sum_{h \in G} h = \sum_{h \in G} gh = N$$

for any $g \in G$.

Theorem 2. Let G be a finite group of order m and A be a G-module such that A is also an R-module where R is any commutative ring with $1/m \in R$. Then N/m is a central idempotent of $R[G]$. Moreover, $H^i(G, A) = 0$ for all $i > 0$ and $H^0(G, A) = eA$ where $e = N/m$.

Proof. We have

$$N^2 = \left(\sum_{g\in G} g\right) N = \sum_{g\in G} gN = \sum_{g\in G} N = mN.$$

Therefore,

$$(N/m)^2 = N^2/m^2 = mN/m^2 = N/m$$

i.e. N/m is an idempotent of $R[G]$. We also have N/m is central since N is central.

Then we have $R[G] = R[G]e \oplus R[G](1-e)$ where $e = N/m$ is an idempotent of $R[G]$. In particular, $R[G]e$ is a projective $R[G]$-module since it is a direct summand of a free module $R[G]$. It follows that $\mathrm{Ext}^i_{R[G]}(R[G]e, A) = 0$ for all $i > 0$.

Since $1/m \in R$, we have that A is naturally a $\mathbb{Z}[1/m][G]$-module. Let $P_\bullet \to \mathbb{Z}$ be a resolution of $\mathbb{Z}$ by free $\mathbb{Z}[G]$-modules. Localizing at m, we obtain $P_\bullet[1/m] \to \mathbb{Z}[1/m]$ which is also a free resolution of $\mathbb{Z}[1/m]$ by $\mathbb{Z}[1/m][G]$-modules.

Since m acts invertibly on A, the universal property of localization gives an isomorphism $\mathrm{Hom}_{\mathbb{Z}[1/m][G]}\big(P_i[1/m], A\big) \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{Z}[G]}(P_i, A)$, $f \mapsto f|_{P_i}$, which is natural in P_i: every $\mathbb{Z}[G]$-linear map $P_i \to A$ extends uniquely to $P_i[1/m]$ by $f(x/m^k) = m^{-k}f(x)$. These isomorphisms commute with the differentials, so

$$\begin{aligned} H^i(G, A) &= H^i\big(\mathrm{Hom}_{\mathbb{Z}[G]}(P_\bullet, A)\big) \\ &\cong H^i\big(\mathrm{Hom}_{\mathbb{Z}[1/m][G]}(P_\bullet[1/m], A)\big) \\ &= \mathrm{Ext}^i_{\mathbb{Z}[1/m][G]}(\mathbb{Z}[1/m], A). \end{aligned}$$

Applying the previous result with $R := \mathbb{Z}[1/m]$, we are done with $\mathbb{Z}[1/m] = R[G]e = \mathbb{Z}[1/m][G]e$ as $\mathbb{Z}[1/m][G]$-modules. Since $e = N/m$, we know $\mathbb{Z}[1/m][G]e = \mathbb{Z}[1/m]e$. By augmentation, we have $\mathbb{Z}[1/m]e \cong \mathbb{Z}[1/m]$ as $\mathbb{Z}[1/m][G]$-modules. Therefore, we have $H^i(G, A) = 0$ for all $i > 0$. For $i = 0$, we have

$$H^0(G, A) = \mathrm{Hom}_{\mathbb{Z}[1/m][G]}(\mathbb{Z}[1/m], A) = \mathrm{Hom}_{\mathbb{Z}[1/m][G]}(\mathbb{Z}[1/m][G]e, A) = eA.$$

□

Corollary 6. Let G be a finite group and A be a G-module which is also a $\mathbb{Z}[1/m]$-module where $m = |G|$. Then $H^i(G, A) = 0$ for all $i > 0$.

Corollary 7. Let G be a finite group. Then $H^0(G, \mathbb{Z}) = \mathbb{Z}, H^1(G, \mathbb{Z}) = 0$, and $H^i(G, \mathbb{Z}) = H^{i-1}(G, \mathbb{Q}/\mathbb{Z})$ for all $i > 1$.

Proof. We have an exact sequence of trivial G-modules $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$. It gives a long exact sequence

$$0 \to H^0(G, \mathbb{Z}) \to H^0(G, \mathbb{Q}) \to H^0(G, \mathbb{Q}/\mathbb{Z}) \to H^1(G, \mathbb{Z}) \to H^1(G, \mathbb{Q}) \to \cdots$$

Since $H^i(G, \mathbb{Q}) = 0$ for all $i > 0$ by the previous corollary, we are done. □

1.3 Bar Resolution

Let G be any discrete group. We want to construct a free resolution of $\mathbb{Z}$ as $\mathbb{Z}[G]$-modules. We set $B_{-1} := \mathbb{Z}$ and the 0-th power of G is the trivial group. For $n \geq 0$, we define B_n to be the free $\mathbb{Z}[G]$-module on G^n. We define the following maps $d_n : B_n \to B_{n-1}$ for $n \geq 0$ (we want the maps to be boundary maps):

$$
\begin{aligned}
&d_0 = \epsilon \quad \text{(augmentation map)},\\
&d_n : B_n \to B_{n-1} \text{ which is defined as}\\
&\quad d_n = \sum_{i=0}^{n} (-1)^i d_n^i,\\
&\quad \text{where } d_n^0([g_1|\cdots|g_n]) = g_1 \cdot [g_2|\cdots|g_n],\\
&\qquad\qquad d_n^i([g_1|\cdots|g_n]) = [g_1|\cdots|g_i g_{i+1}|\cdots|g_n] \text{ for } 1 \leq i \leq n-1,\\
&\qquad\qquad d_n^n([g_1|\cdots|g_n]) = [g_1|\cdots|g_{n-1}].
\end{aligned}
$$

The maps d_n are boundary maps because of their simplicial structure (look at the Dold–Kan correspondence).

Theorem 3. The sequence $B_\bullet$ is exact.

Proof. We shall show that $\cdots \to B_n \xrightarrow{d_n} B_{n-1} \to \cdots \to B_0 \xrightarrow{d_0} B_{-1} \to 0$ is exact by constructing a null-homotopy. We want to construct maps $h_n : B_n \to B_{n+1}$ such that $d_{n+1} \circ h_n + h_{n-1} \circ d_n = \mathrm{id}_{B_n}$ for all $n \geq -1$. Define $h_n : B_n \to B_{n+1}$ as

$$h_n(g_0 \cdot [g_1|\cdots|g_n]) = [g_0|g_1|\cdots|g_n].$$

For $n = -1$, it is clear that $d_0 \circ h_{-1} = \mathrm{id}_{B_{-1}}$. Then when $n \geq 1$, we have

$$
\begin{aligned}
d_{n+1} \circ h_n([g_1|\cdots|g_n]) &= d_{n+1}([e|g_1|\cdots|g_n])\\
&= [g_1|\cdots|g_n] - [g_1|\cdots|g_n]\\
&\quad + \sum_{i=1}^{n-1} (-1)^{i+1}[e|g_1|\cdots|g_i g_{i+1}|\cdots|g_n] + (-1)^{n+1}[e|\cdots|g_{n-1}]
\end{aligned}
$$

and

$$\begin{aligned} h_{n-1} \circ d_n([g_1|\cdots|g_n]) &= h_{n-1}(g_1 \cdot [g_2|\cdots|g_n] \\ &\quad + \sum_{i=1}^{n-1}(-1)^i[g_1|\cdots|g_ig_{i+1}|\cdots|g_n] + (-1)^n[g_1|\cdots|g_{n-1}]) \\ &= [g_1|\cdots|g_n] + \sum_{i=1}^{n-1}(-1)^i[e|g_1|\cdots|g_ig_{i+1}|\cdots|g_n] \\ &\quad + (-1)^n[g_1|\cdots|g_{n-1}]. \end{aligned}$$

Therefore, we have

$$d_{n+1} \circ h_n([g_1|\cdots|g_n]) + h_{n-1} \circ d_n([g_1|\cdots|g_n]) = [g_1|\cdots|g_n].$$

For $n = 0$, we have

$$\begin{aligned} d_1 \circ h_0([g_1]) = d_1([e|g_1]) = [g_1] - [e], \\ h_{-1} \circ d_0([g_1]) = h_{-1}(\epsilon([g_1])) = h_{-1}([e]) = [e]. \end{aligned}$$

Therefore, we have $d_{n+1} \circ h_n + h_{n-1} \circ d_n = \mathrm{id}_{B_n}$ for all $n \geq -1$. □

By the previous theorem, we have a free resolution of $\mathbb{Z}$ as $\mathbb{Z}[G]$-modules. This is called the *bar resolution* of $\mathbb{Z}$ as $\mathbb{Z}[G]$-modules invented by Eilenberg and MacLane.

We also observe that $\mathrm{Hom}_{\mathbb{Z}[G]}(B_n, B_{n-1}) \cong \mathrm{Hom}_{\mathsf{Set}}(G^n, B_{n-1})$ because B_n is a free $\mathbb{Z}[G]$-module on G^n.

Let's consider a special case: $B_2 \xrightarrow{d_2} B_1 \xrightarrow{d_1} B_0 \xrightarrow{d_0} B_{-1} \to 0$. Let A be a G-module. Then we have

$$H^i(G, A) = H^i(\mathrm{Hom}_{\mathsf{Set}}(G^\bullet, A)).$$

For $G^0 = \{e\}$, we have $\mathrm{Hom}_{\mathsf{Set}}(G^0, A) \cong A$. We consider the pullback map $d_1^* : \mathrm{Hom}_{\mathsf{Set}}(G^0, A) \to \mathrm{Hom}_{\mathsf{Set}}(G^1, A)$ of d_1 which is given by $d^1(g) = g - 1$. So we have $H^0(G, A) = \ker(d_1^*) = \{a \in A \ : \ g \cdot a - a = 0\} = A^G$.

We also have $d_2(f)(g_1, g_2) = g_1 \cdot f(g_2) - f(g_1 g_2) + f(g_1)$. Hence $Z^1(G, A) = \ker(d_2^*) = \{f : G \to A \ : \ g_1 \cdot f(g_2) - f(g_1 g_2) + f(g_1) = 0\}$. Such functions f are called *crossed homomorphisms*. In addition we have $B^1(G, A) = \mathrm{im}(d_1^*) = \{f : G \to A \ : \ f(g) = g \cdot a - a \text{ for some } a \in A\}$. Any map in the boundary is also a crossed homomorphism because

$$g_1 \cdot f(g_2) - f(g_1 g_2) + f(g_1) = g_1 \cdot (g_2 \cdot a - a) - (g_1 g_2 \cdot a - a) + (g_1 \cdot a - a) = 0.$$

And such maps are called *principal crossed homomorphisms*. And $H^1(G, A) = Z^1(G, A)/B^1(G, A)$ is the group of crossed homomorphisms modulo principal crossed homomorphisms.

Corollary 8. Let A be a trivial G-module. Then, we have

$$H^0(G, A) \cong A,$$
$$H^1(G, A) \cong \mathrm{Hom}_{\mathsf{group}}(G, A).$$

Proof. We have $Z^1(G, A) = \mathrm{Hom}_{\mathsf{group}}(G, A)$ and $B^1(G, A) = 0$ since A is a trivial G-module. □

We already looked at the case of $A = \mathbb{Z}$ of the following corollary.

Corollary 9. Suppose G is a finite group and A is a torsion-free trivial G-module. Then, we have $H^1(G, A) = 0$.

Theorem 4. Suppose G is a finite group of order m, and let A be a G-module. Then $H^n(G, A)$ is annihilated by m for all $n > 0$, i.e., $H^n(G, A)$ is a $\mathbb{Z}/m\mathbb{Z}$-module for all $n > 0$.

Proof. Our strategy to prove this is to use null-homotopy again. Let $B_\bullet \xrightarrow{\epsilon} \mathbb{Z}$ be the bar resolution of $\mathbb{Z}$. Let's consider the map of the complexes $B_\bullet \to B_\bullet$ given by where $N = \sum_{g \in G} g$ is the norm element. We want to show that this is null-homotopic in positive degrees. Let $\nu_n : B_n \to B_{n+1}$ be the map given by

$$\nu_n([g_1|g_2|\cdots|g_n]) = \epsilon_n \sum_{g \in G} [g_1|\cdots|g_n|g]$$

$$\begin{array}{ccccccc} \cdots & \longrightarrow & B_2 & \longrightarrow & B_1 & \longrightarrow & B_0 \\ & & \downarrow{\scriptstyle m} & & \downarrow{\scriptstyle m} & & \downarrow{\scriptstyle m-N} \\ \cdots & \longrightarrow & B_2 & \longrightarrow & B_1 & \longrightarrow & B_0 \end{array}$$

where $\epsilon_n = (-1)^{n+1}$. We claim that $\nu_{n-1} \circ d_n + d_{n+1} \circ \nu_n = m$ in all positive degrees. Let $\omega = [g_1|\cdots|g_n] \in B_n$. Then, we have

$$\begin{aligned} d_{n+1} \circ \nu_n(\omega) &= \epsilon_n \sum_{g\in G} d_{n+1}([g_1|\cdots|g_n|g]) \\ &= (-1)^{n+1} \sum_{g\in G} (g_1[g_2|\cdots|g_n|g]) \\ &\quad + (-1)^{n+1} \sum_{g\in G} \left(\sum_{i=1}^{n-1} (-1)^i [g_1|\cdots|g_i g_{i+1}|\cdots|g_n|g] \right) \\ &\quad + (-1)^{n+1} \sum_{g\in G} ((-1)^n [g_1|\cdots|g_n g]) + \sum_{g\in G} ([g_1|\cdots|g_n]), \\ \nu_{n-1} \circ d_n(\omega) &= \nu_{n-1} \left(g_1[g_2|\cdots|g_n] + \sum_{i=1}^{n-1} (-1)^i [g_1|\cdots|g_i g_{i+1}|\cdots|g_n] + (-1)^n [g_1|\cdots|g_{n-1}] \right) \\ &= (-1)^n \sum_{g\in G} g_1[g_2|\cdots|g_n|g] + (-1)^n \sum_{g\in G} \left(\sum_{i=1}^{n-1} (-1)^i [g_1|\cdots|g_i g_{i+1}|\cdots|g_n|g] \right) \\ &\quad + (-1)^n \sum_{g\in G} ([g_1|g_2|\cdots|g_{n-1}|g]). \end{aligned}$$

We observe that

$$\sum_{g\in G} ([g_1|g_2|\cdots|g_{n-1}|g]) = \sum_{g\in G} ([g_1|\cdots|g_n g])$$

since g runs over all elements of G. Therefore, from the above, we have

$$d_{n+1} \circ \nu_n(\omega) + \nu_{n-1} \circ d_n(\omega) = \sum_{g\in G} [g_1|\cdots|g_n] = m(\omega).$$

For the 0-th degree, we have

$$d_1 \circ \nu_0([]) = d_1\left(-\sum_{g\in G}[g]\right) = -\sum_{g\in G}[g] + \sum_{g\in G}[1] = m - N.$$

Thus, $m \cdot H^n(G, A) = 0$ for all $n > 0$. □

Corollary 10. If A is $\mathbb{Z}[1/m]$-module, then $H^n(G, A) = 0$ for all $n > 0$.

Proof. Since A is a $\mathbb{Z}[1/m]$-module, we have m is an automorphism for A; hence an automorphism for $H^n(G, A)$ for all $n > 0$. But $m \cdot H^n(G, A) = 0$ for all $n > 0$ by the previous theorem. Therefore, $H^n(G, A) = 0$ for all $n > 0$. □

We continue to assume that G is a finite group of order m. By averaging over G, we get a trivial G-module structure on a $\mathbb{Z}[1/m][G]$-module A.

Definition 3. The Reynolds operator $R : A \to A^G$ is defined by

$$R(a) = \frac{1}{m}\sum_{g\in G} g \cdot a.$$

1.4 Periodic Cohomology

1.4.1 Group Cohomology of a Cyclic Group

Now we will compute the cohomology of a group in a simpler case, when the group is cyclic. Let G be a cyclic group of order n generated by σ.

Before beginning the computation, we observe the following:

Remark 4.

1. $\mathbb{Z}[\mathbb{Z}] = \mathbb{Z}[t, t^{-1}]$.
2. $\mathbb{Z}[\mathbb{Z}_{>0}] = \mathbb{Z}[t]$.
3. $\mathbb{Z}[\mathbb{Z}/n\mathbb{Z}] = \mathbb{Z}[t]/(t^n - 1)$.

So we now know that the group ring $\mathbb{Z}[G]$ is isomorphic to the quotient ring $\mathbb{Z}[t]/(t^n-1)$ via the map $\sigma \mapsto t$. Let N be the norm element $N = 1+\sigma+\cdots+\sigma^{n-1} \in \mathbb{Z}[G]$. In $\mathbb{Z}[t]/(t^n-1)$, we have $N = 1+t+\cdots+t^{n-1}$. So we observe that $N(\sigma-1)$ is corresponding to $N(t-1) = (1+t+\cdots+t^{n-1})(t-1) = t^n - 1 = 0$ in $\mathbb{Z}[t]/(t^n-1)$. It follows that $\mathrm{im}(N) \subseteq \ker(\sigma-1)$ and $\mathrm{im}(\sigma-1) \subseteq \ker(N)$ when we consider N and $\sigma - 1$ as multiplication maps on $\mathbb{Z}[G]$. Moreover, we have

Proposition 3. $\ker(N) = \mathrm{im}(\sigma - 1)$ and $\ker(\sigma - 1) = \mathrm{im}(N)$.

Proof. Suppose that $a \in \ker(N)$, i.e., $Na = 0$. Let $a(t)$ be the polynomial in $\mathbb{Z}[t]$ corresponding to a. Then, we have $N(t)a(t) = (t^n-1)b(t)$ for some $b(t) \in \mathbb{Z}[t]$. Since $N(t)(t-1) = t^n - 1$, we have $a(t) = (t-1)b(t)$ in $\mathbb{Z}[t]$ and hence $a \in \mathrm{im}(\sigma - 1)$; hence $\ker(N) = \mathrm{im}(\sigma-1)$. With the same argument, we can show that $\ker(\sigma-1) = \mathrm{im}(N)$. □

With the property above, we can construct a free resolution of $\mathbb{Z}$ as a $\mathbb{Z}[G]$-module which is more convenient than the bar resolution. We call this the 2-*periodic resolution* of $\mathbb{Z}$ as a $\mathbb{Z}[G]$-module.

Corollary 11. The following chain is exact:

$$\cdots \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where ϵ is the augmentation map.

Suppose now that A is a G-module. Let $B^\bullet \xrightarrow{\epsilon} \mathbb{Z}$ be the 2-periodic resolution of $\mathbb{Z}$ as a $\mathbb{Z}[G]$-module. Then, we have

$$\begin{aligned} &H^i(G, A) = H^i(\mathrm{Hom}_{\mathbb{Z}[G]}(B^\bullet, A)), \\ &\mathrm{Hom}_{\mathbb{Z}[G]}(B^\bullet, A) = \cdots \xleftarrow{N} A \xleftarrow{\sigma-1} A \xleftarrow{N} A \xleftarrow{\sigma-1} A. \end{aligned}$$

Thus, we get

$$\begin{aligned} H^0(G, A) &= \ker(\sigma - 1) = A^G, \\ H^{2i+1}(G, A) &= {}_N A/(\sigma - 1)A, \\ H^{2i+2}(G, A) &= A^G/N(A) \end{aligned}$$

where ${}_N A = \{a \in A : Na = 0\}$ and $i \geq 0$. We can see that the cohomology groups of a cyclic group are periodic with period 2. Here are some applications of this computation.

1.4.2 Hilbert 90

Let L/K be a finite Galois extension of fields with Galois group $G = \mathrm{Gal}(L/K)$. What are the Galois cohomology groups of G with coefficients in L or $L^\times$? For the 0-th cohomology group, we can easily see that $H^0(G, L) = L^G = K$ and $H^0(G, L^\times) = (L^\times)^G = K^\times$. How about the higher cohomology groups?

Theorem 5 (Hilbert's Theorem 90)**.** $H^1(G, L^\times) = 0$.

Before proving this theorem, we prove the following lemma.

Lemma 4 (Linear independence of automorphisms)**.** Let E be a field and $\sigma_1, \cdots, \sigma_n$ be distinct automorphisms of E. Then, for any $c_1, \cdots, c_n \in E$, $\sum_{i=1}^n c_i\sigma_i(x) = 0$ for all $x \in E$ implies $c_1 = c_2 = \cdots = c_n = 0$.

Proof. We will prove this by induction on n. We may assume that none of c_i is zero, otherwise we can remove the corresponding σ_i and reduce to a smaller n. If $n = 1$, then we have $c_1\sigma_1(1) = c_1 \neq 0$.

Now we consider the case $n > 1$ and the statement is true for $n - 1$, but suppose that $\sum_{i=1}^n c_i\sigma_i = 0$ for the sake of contradiction. We can set $c_n = -1$ without loss of generality. Let $x, y \in E$. We have $\sigma_n(x) = \sum_{i=1}^{n-1} c_i\sigma_i(x)$, and $\sigma_n(x)\sigma_n(y) = \sigma_n(xy)$.

The RHS is equal to $\sum_{i=1}^{n-1} c_i\sigma_i(xy) = \sum_{i=1}^{n-1} c_i\sigma_i(x)\sigma_i(y)$ while the LHS is equal to $\sum_{i=1}^{n-1} c_i\sigma_i(x)\sigma_n(y)$. It follows that

$$0 = \sum_{i=1}^{n-1} c_i\sigma_i(x)\sigma_n(y) - \sum_{i=1}^{n-1} c_i\sigma_i(x)\sigma_i(y) = \sum_{i=1}^{n-1} c_i\sigma_i(x)(\sigma_n(y) - \sigma_i(y)).$$

By the induction hypothesis, we have $c_i(\sigma_n(y) - \sigma_i(y)) = 0$ for all i. We also have $c_i \neq 0$ for all i, so we must have $\sigma_n(y) - \sigma_i(y) = 0$ for all i. Recalling that y is arbitrary, we have $\sigma_n = \sigma_i$ for all i which is a contradiction. □

Proof of Theorem 5. Recall that $H^1(G, L^\times)$ is the group of crossed homomorphisms modulo principal crossed homomorphisms. Let $\alpha : G \to L^\times$ be a crossed homomorphism. For $g, h \in G$, we have $\alpha(gh) = g\alpha(h) \cdot \alpha(g)$. We note that each $g \in G$ is an automorphism of $L^\times$, and none of $\alpha(g)$ is zero. The lemma implies that there exists $x \in L^\times$ such that $\beta := \sum_{g\in G} \alpha(g)g(x) \neq 0$. For $h \in G$, we have

$$\begin{aligned} h(\beta) &= h\left(\sum_{g\in G} \alpha(g)g(x)\right) = \sum_{g\in G} h(\alpha(g))hg(x) \\ &= \sum_{g\in G} \alpha(h)^{-1}\alpha(hg)hg(x) = \alpha(h)^{-1}\sum_{g\in G} \alpha(hg)hg(x) = \alpha(h)^{-1}\beta. \end{aligned}$$

So we have $\alpha(h) = \beta/h(\beta)$ for all $h \in G$. Therefore, α is a principal crossed homomorphism, and we conclude that $H^1(G, L^\times) = 0$. □

Let's take a look at the original version of Hilbert 90.

Theorem 6 (Hilbert's Theorem 90, Original Version)**.** Let L/K be a cyclic Galois extension of fields with Galois group $G = \mathrm{Gal}(L/K)$ generated by σ. Let $N : L^\times \to K^\times$ be the norm map given by $N(x) = x\sigma(x)\cdots\sigma^{n-1}(x)$. Then, for any $x \in L^\times$ with $N(x) = 1$, there exists $y \in L^\times$ such that $x = y/\sigma(y)$.

Proof. We want to show that Theorem 5 implies the original version of Hilbert 90. Recall that $H^1(G, L^\times) = \ker(N)/\mathrm{im}(\sigma - 1)$ because G is cyclic. So if $N(x) = 1$, then $x \in \ker(N)$, and hence $x \in \mathrm{im}(\sigma - 1)$ by Theorem 5. Therefore, there exists $y \in L^\times$ such that $x = y/\sigma(y)$. □

1.4.3 Tate Cohomology

This is the main application of the 2-periodic resolution. Let G be a cyclic group acting on an abelian group A.

Definition 4 (Tate Cohomology)**.** The Tate cohomology groups $\hat{H}^n(G, A)$ are defined as follows:

$$\hat{H}^n(G, A) = \begin{cases} A^G/N(A) & n \text{ is even,} \\ {}_NA/(\sigma - 1)A & n \text{ is odd.} \end{cases}$$

So now we can consider the negative indices. In fact, we can do this for groups that are not necessarily cyclic. We define the negative part with the interaction of cohomology and homology.

Let $\alpha : R \to S$ be a ring homomorphism such that S is a free-R-module of rank $n < \infty$. Let M, P be S-modules. Then we have the corestriction map $\mathrm{Cores}_R^S : \mathrm{Ext}_R^i(P, M) \to \mathrm{Ext}_S^i(P, M)$ and the restriction map $\mathrm{Res}_R^S : \mathrm{Ext}_S^i(P, M) \to \mathrm{Ext}_R^i(P, M)$.

Proposition 4. The composition $\mathrm{Cores}_R^S \circ \mathrm{Res}_R^S$ is multiplication by n.

Proof. First, we start with the case $i = 0$. We have the following diagram:

$$\begin{array}{ccccc} \mathrm{Hom}_S(P, M) & \xhookrightarrow{\mathrm{Res}_R^S} & \mathrm{Hom}_R(P, M) & \xrightarrow{\sim} & \mathrm{Hom}_S(P \underset{R}{\otimes} S, M) \\ & & & & \downarrow \wr \\ \mathrm{Hom}_S(P, M) & \xleftarrow{\Delta^*} & \bigoplus \mathrm{Hom}_S(Pe_i, M) & \xleftarrow{\sim} & \mathrm{Hom}_S(P^n, M) \end{array}$$

where $e_1, \ldots, e_n$ is a basis of S as a free-R-module. We have $\Delta^*(f_1, \ldots, f_n) = \sum_{i=1}^n f_i$. We observe that the composition gives $f \mapsto (f, \ldots, f) \mapsto \sum_{i=1}^n f = nf$.

It gives a natural transformation $\eta : \mathrm{Hom}_S(P, -) \to \mathrm{Hom}_S(P, -)$ on S-Mod such that $\eta(M)$ is multiplication by n. Since $\mathrm{Hom}_S(P, -)$ is a left exact functor, we can extend η to a natural transformation $\eta^i : \mathrm{Ext}_S^i(P, -) \to \mathrm{Ext}_S^i(P, -)$ for all $i \geq 0$, and it must be multiplication by n as well. □

Corollary 12. Let G be a discrete group and H be a subgroup of a finite index n. Let $A \in G$-Mod. Then the composition

$$H^i(G, A) \xrightarrow{\mathrm{Res}_H^G} H^i(H, A) \xrightarrow{\mathrm{Cores}_H^G} H^i(G, A)$$

is multiplication by $n = |G/H|$.

This can be interpreted as the algebraic version of a topological fact. Let X be a connected topological space and $Y \to X$ be an n-sheeted covering space. Then the restriction map is corresponding to the pullback p^* and the corestriction map is corresponding to the transfer map $p_!$. (So we sometimes call the corestriction map the transfer map.) Copying a class to n sheets and adding them back up gives n times the original class.

Corollary 13. Suppose G is a finite group of order n and A is a G-module. Then $H^i(G, A)$ is annihilated by n for all $i > 0$.

Proof. Apply the previous corollary with the trivial subgroup $H = \{e\}$. Then we observe $H^i(G, A) \to H^i(\{e\}, A) \to H^i(G, A)$ is multiplication by n. But $H^i(\{e\}, A) = 0$ for all $i > 0$, so we conclude that $H^i(G, A)$ is annihilated by n for all $i > 0$. □

Let G be a discrete group and H be a normal subgroup of finite index, and A be a G-module. Then we have the inflation map, the